

① $X \equiv 1 \pmod{3}, X \equiv 2 \pmod{5}, X \equiv 3 \pmod{7}$

product $N = 3 \cdot 5 \cdot 7 = 105$

Compute Contributions:

$N_1 = 105/3 = 35, 35 \equiv 2 \pmod{3}$, inverse of 2 mod 3 is 2. Term: $1 \cdot 35 \cdot 2 = 70$

$N_2 = 105/5 = 21, 21 \equiv 1 \pmod{5}$, inverse 1.

Term: $2 \cdot 21 \cdot 1 = 42$

$N_3 = 105/7 = 15, 15 \equiv 1 \pmod{7}$, inverse 1

Term: $3 \cdot 15 \cdot 1 = 45$

Sum = $70 + 42 + 45 = 157$ reduce mod 105:

$$157 \equiv 52 \pmod{105}$$

$$\therefore X \equiv 52 \pmod{105}$$

② $X \equiv 5 \pmod{11}, X \equiv 14 \pmod{29}, X \equiv 15 \pmod{31}$

product $N = 11 \cdot 29 \cdot 31 = 9889$

Compute Contribution:

$$N_1 = 9889/11 = 899, 899 \equiv 8 \pmod{11},$$

$$\text{inverse of } 8 \pmod{11} \text{ is } 7. \text{ Term: } 5 \cdot 899 \cdot 7 \\ = 31465$$

$$N_2 = 9889/29 = 341, 341 \equiv 22 \pmod{29}, \text{ inverse} \\ \text{of } 22 \pmod{29} \text{ is } 4. \text{ Term: } 14 \cdot 341 \cdot 4 = 19096$$

$$N_3 = 9889/31 = 319, 319 \equiv 9 \pmod{31}, \text{ inverse} \\ \text{of } 9 \pmod{31} \text{ is } 7. \text{ Term: } 15 \cdot 319 \cdot 7 = 33495$$

$$\text{Sum} = 31465 + 19096 + 33495 = 84056$$

$$\text{Reduce mod } 9889: 84056 = 4944 \pmod{9889}$$

$$X \equiv 4944 \pmod{9889}$$

$$\textcircled{c} X \equiv 5 \pmod{6}, X \equiv 4 \pmod{11}, X \equiv 3 \pmod{17}.$$

$$\text{product } N = 6 \cdot 11 \cdot 17 = 1122$$

Complete Contribution:

$$N_1 \equiv 1122/6 = 187, 187 \equiv 1 \pmod{6}, \text{inverse } 1$$

$$\text{Term} : 5 \cdot 187 \cdot 1 = 935$$

$$N_2 \equiv 1122/11 = 102, 102 \equiv 3 \pmod{11}, \text{inverse of } 3$$

$$\text{mod } 11 \text{ is } 4, \text{Term} : 4 \cdot 102 \cdot 4 = 1632$$

$$N_3 \equiv 1122/17 = 66, 66 \equiv 15 \pmod{17}, \text{inverse of } 15$$

$$\text{mod } 17 \text{ is } 8. \text{Term} : 3 \cdot 66 \cdot 8 = 1584$$

$$\text{Sum} = 935 + 1632 + 1584 = 4151$$

$$\text{Reduce mod } 1122 : 4151 \equiv 785 \pmod{1122}$$

$$X \equiv 785 \pmod{1122}.$$